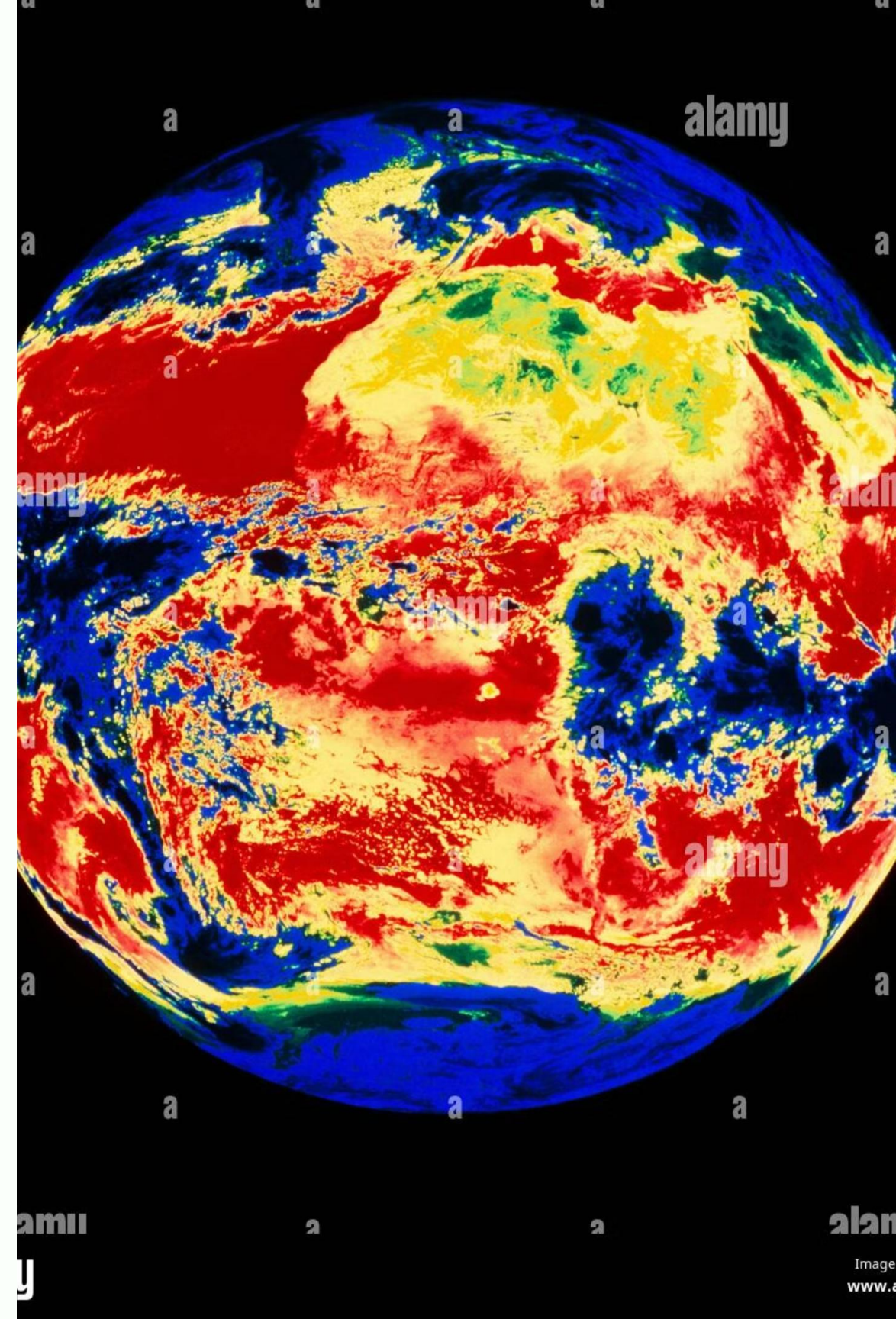


Spatio-Temporal Gap Filling of Land Surface Temperature (LST).

Hybrid Approach: Interpolation + Machine Learning | Dataset: LST
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Problem Statement

Satellite-derived Land Surface Temperature (LST) datasets frequently contain **missing or corrupted values**, compromising downstream climate and urban heat analyses.

Cloud Cover

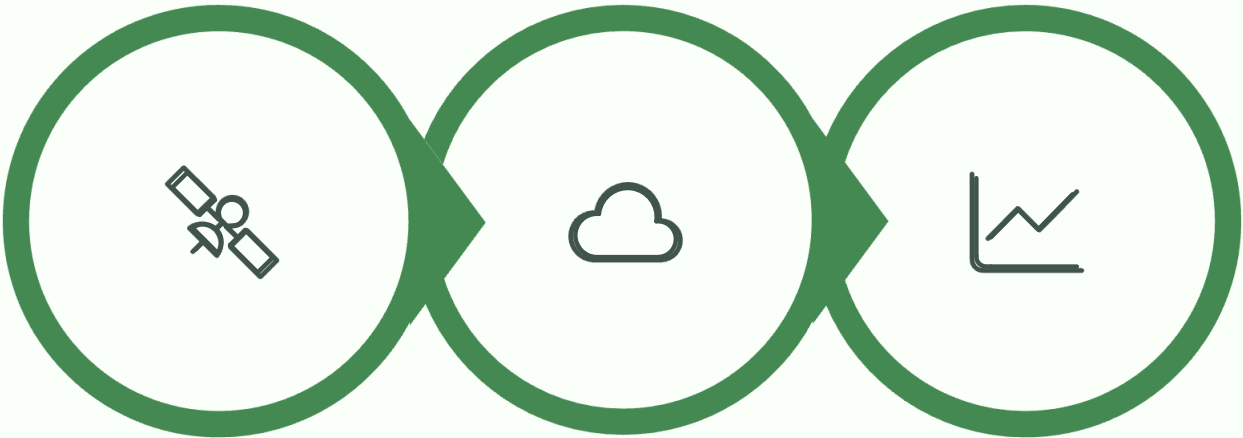
Clouds obstruct thermal sensors, rendering large spatial extents unobservable.

Sensor Limitations

Hardware malfunctions and orbital gaps introduce systematic data voids.

Impact

Incomplete LST grids undermine climate modeling, urban heat island studies, and land use analyses.



*Satellite
Acquisition*

Missing Values

*Degraded
Analysis*

Research Objectives

This study aims to reconstruct spatially and temporally coherent LST fields from incomplete satellite observations using a hybrid computational framework.

1

Reconstruct Missing LST Values

Recover gapped pixels to produce complete, analysis-ready raster grids.

2

Leverage Spatial Context

Exploit neighborhood pixel relationships through interpolation techniques.

3

Incorporate Temporal Patterns

Use multi-date LST composites to capture seasonal and diurnal variability.

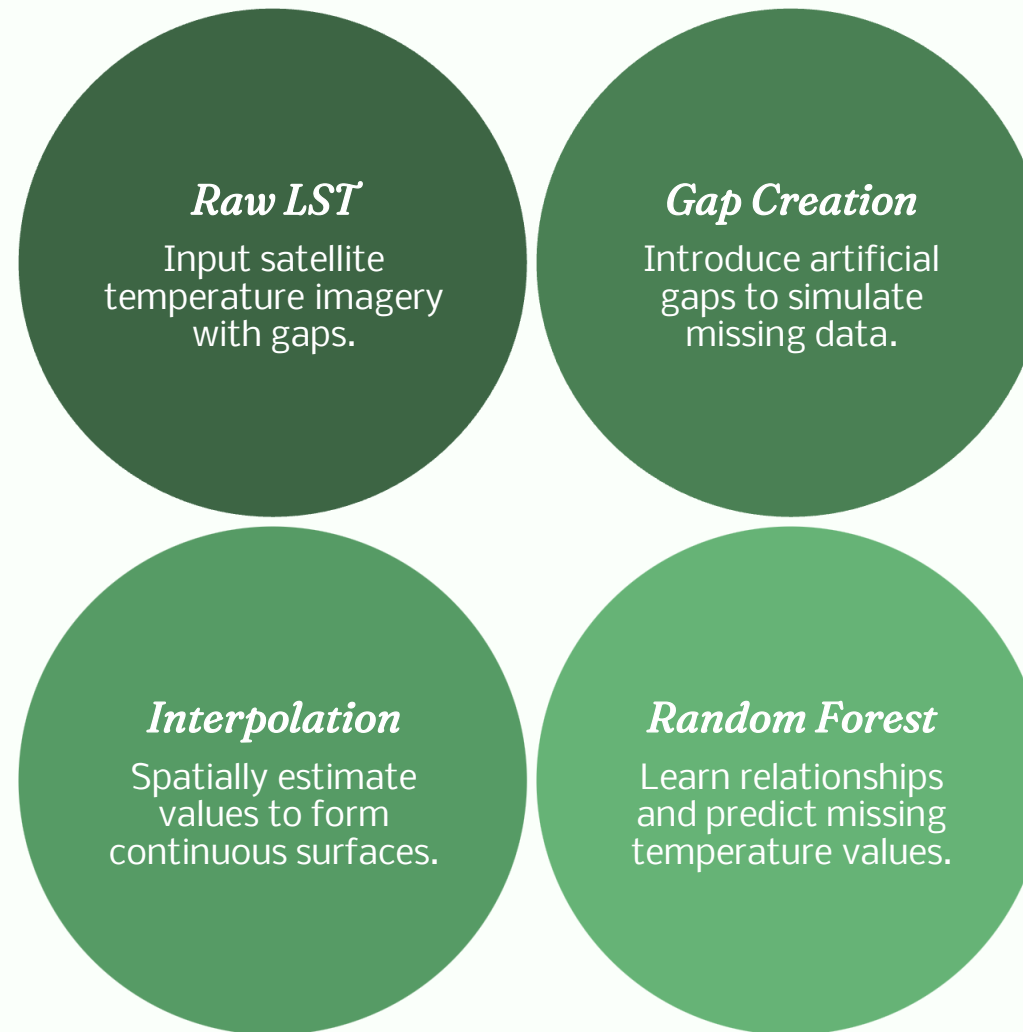
4

Apply Machine Learning

Deploy Random Forest regression to refine and generalize reconstructed values.

Methodology Overview

A five-stage pipeline transforms raw, gap-ridden LST imagery into complete reconstructed outputs through sequential spatial and machine learning operations.



Each stage feeds directly into the next – interpolated surfaces serve as the primary feature input for the Random Forest model, ensuring spatial coherence is preserved throughout.

EXPERIMENTAL DESIGN

Artificial Gap Creation

To rigorously evaluate reconstruction performance, **artificial gaps were introduced** into known complete LST scenes via random pixel masking – enabling direct comparison of predicted vs. observed values.

10% Masking

Low-density gaps; representative of partly cloudy conditions.

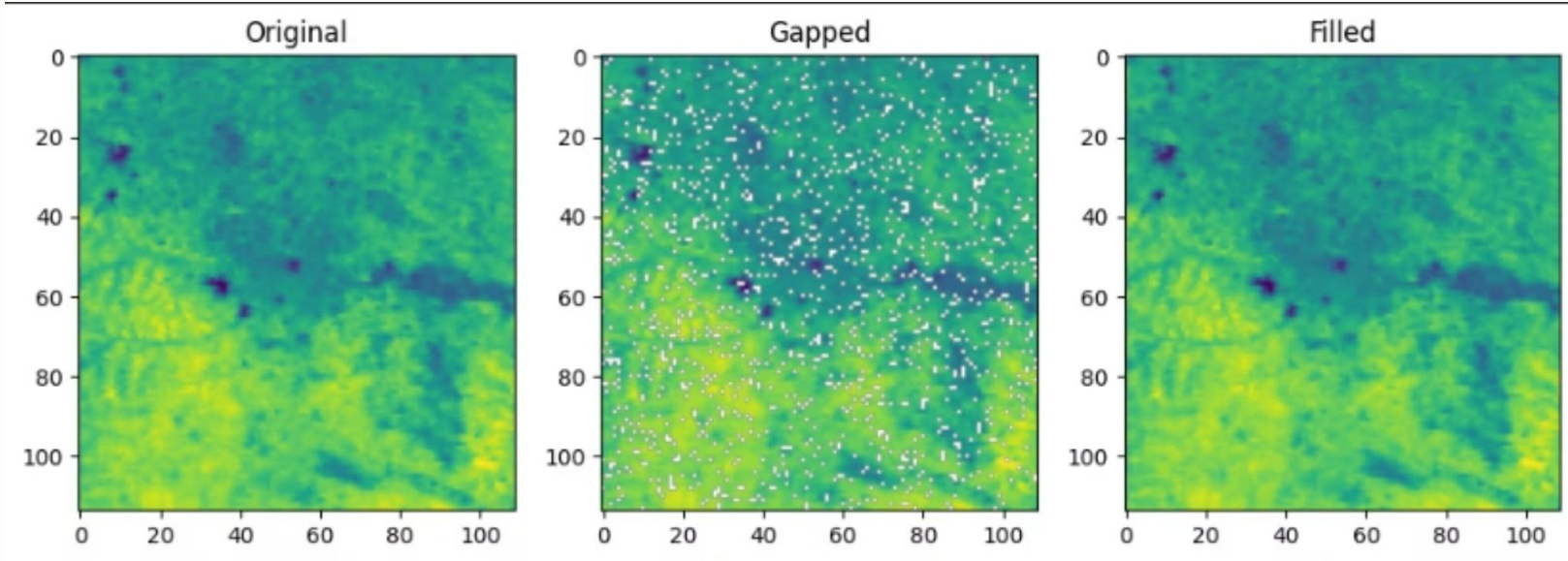
25% Masking

Moderate sparsity; reflects common mid-latitude cloud climatology.

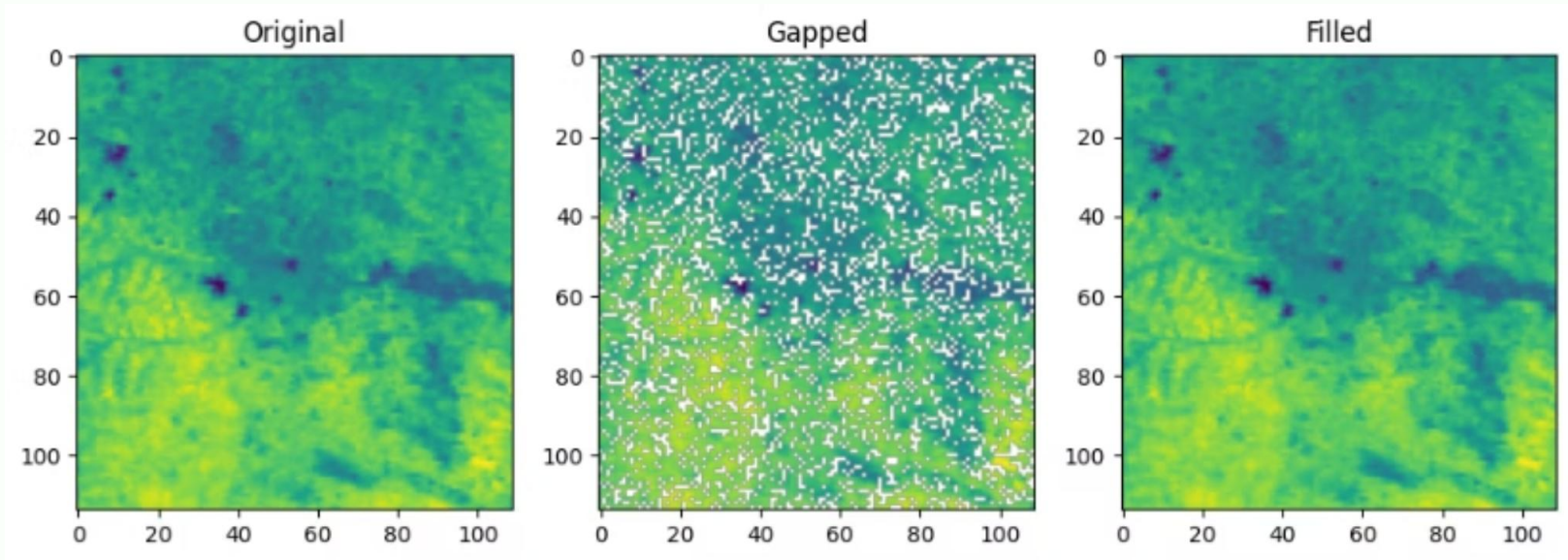
50% Masking

Severe data loss; stress-tests model robustness under worst-case scenarios.

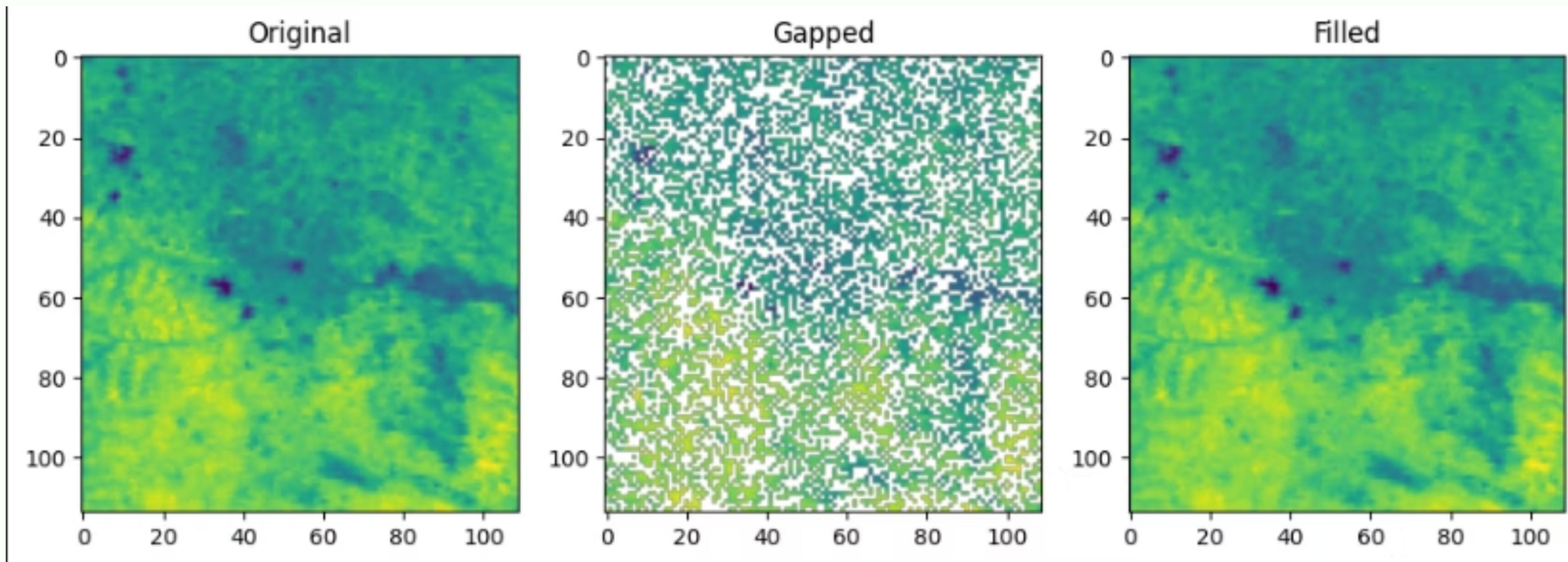
Before and After Gap Filling



With 10% Random Gaps

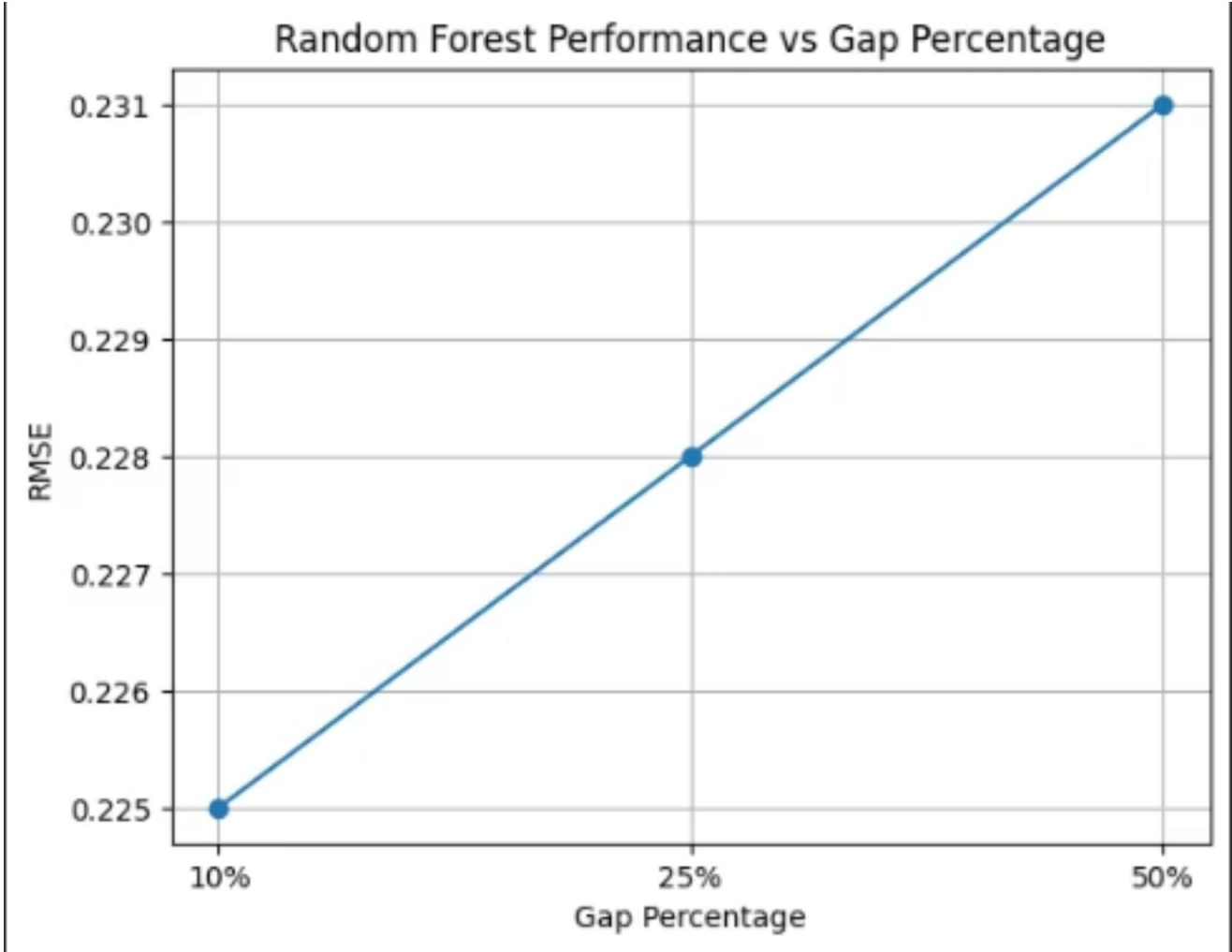


With 25% Random Gaps



With 50% Random Gaps

RMSE Comparison



RMSE Summary by Gap Level

Gap Level	RMSE (°C)
10%	0.225
25%	0.228
50%	0.231

RMSE increases marginally across all gap scenarios – a variation of just **0.006°C** from 10% to 50% masking – demonstrating strong model stability under increasing data scarcity.